



OPEN Dynamic solar irradiance estimation for vehicle thermal management using a multi-modal machine learning framework

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Accurate solar irradiance estimation for moving vehicles is essential for effective thermal management and energy optimization, particularly in electric vehicles (EVs) where cabin climate control directly impacts battery usage and driving range. However, existing approaches often overlook the dynamic environmental conditions encountered during vehicle movement, limiting their applicability in real-world scenarios. This study proposes a novel machine learning framework that integrates satellite remote sensing and sky camera imagery to estimate solar irradiance around moving vehicles in real time. The framework was validated using comprehensive field data collected from an instrumented vehicle equipped with solar irradiance sensors, GPS tracking, and environmental monitoring systems across various routes and seasonal conditions. By fusing multi-source environmental data, the model delivers high spatiotemporal resolution predictions of solar irradiance under diverse urban and atmospheric conditions. Results show strong performance, with normalized root-mean-square errors of 14.61% in summer and 17.10% in winter. These accurate predictions enable adaptive, location-aware thermal management strategies that proactively adjust climate control based on anticipated solar exposure. The integration of satellite and ground-based data sources ensures scalability across regions and vehicle types, establishing a foundation for intelligent thermal management systems that enhance energy efficiency and passenger comfort in both conventional and electric vehicles.

Keywords Moving vehicles, Machine learning, Solar irradiance, Satellite data, Sky images

Abbreviations

CatBoost	Categorical boosting
DHI	Diffuse horizontal irradiance, W/m ²
DNI	Direct normal irradiance, W/m ²
EV	Electric vehicle
GHI	Global horizontal irradiance, W/m ²
LightGBM	Light gradient boosting machine
MBE	Mean bias error
nMBE	Normalized mean bias error
RMSE	Root-mean-square error
nRMSE	Normalized root-mean-square error
SHAP	Shapley additive explanations
XGBoost	Extreme gradient boosting

List of symbols

AST	Apparent solar time
D_{es}	Sun to earth distance
E_a	Extraterrestrial irradiance, W/m ²
L_t	Linked turbidity
n	Number of samples
P_{sat}	Pixel value or digital number from satellite
RH_{sat}	Relative humidity from satellite

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T_{sat}	Temperature from satellite
θ	Solar zenith angle, °
y_m	Actual data
y_e	Estimated data
$\overline{y_m}$	Mean value of actual data
γ	Sun-blocking information

Vehicles represent a critical component of global transportation systems, and their interaction with the surrounding environment significantly impacts passenger comfort, energy efficiency, and system design. Among these environmental factors, solar irradiance plays a key role in influencing cabin thermal conditions^{1,2}, especially when vehicles are exposed to direct sunlight. When parked or operated in sunny conditions, cabin temperatures can rise rapidly, leading to passenger discomfort and increased energy demand from cooling systems³. This issue is especially critical in hot climates or during summer months, where managing internal heat loads becomes essential for maintaining comfort and minimizing energy consumption.

Research highlights that 50% to 75% of the thermal energy accumulated inside a vehicle cabin is a result of solar energy transmitted and absorbed through the glass window^{4,5}. The effect is particularly pronounced in modern vehicle designs, especially electric vehicles (EVs), which often feature larger glass areas such as panoramic roofs and wide windshields to enhance visibility and aesthetics. While these design choices improve occupant experience, they also increase the exposure to solar radiation, intensifying thermal gains during summer. However, during colder seasons, solar irradiance can act as a passive heat source, reducing reliance on electric or fuel-based heating systems⁶. This dual nature highlights the importance of accurately estimating solar irradiance to support intelligent climate control strategies in all vehicle types, with even greater importance for EVs, where cabin thermal management directly affects battery performance and driving range⁷.

For this reason, solar irradiance data is crucial for understanding and addressing the effects of solar heat gain inside a vehicle cabin. Accurate solar irradiance measurements enable manufacturers to develop effective solutions, such as advanced glazing materials⁸ and enhanced insulation, to minimize thermal loads⁹. Solar irradiance including direct and diffuse irradiance data also plays a critical role in predictive thermal modeling^{10,11}, enabling engineers to simulate how different vehicle designs will perform under various climate conditions before physical prototypes are built. Furthermore, accurate solar irradiance information is valuable for photovoltaic applications^{12,13}, supporting the optimization of energy harvesting and integration of solar panels on vehicle surfaces, which can enhance overall vehicle efficiency. However, measuring these components typically involves equipment with moving parts, such as a sun tracker, which is challenging to install on a vehicle^{14,15}. Therefore, as a practical alternative, solar irradiance models provide an effective solution for estimating these values without requiring complex instrumentation.

Recent advancements in satellite remote sensing technology have significantly improved the accuracy of solar irradiance estimation, offering high-resolution spatial and temporal coverage that is crucial for assessing solar energy potential across large areas^{16–18}. However, while satellite imagery provides valuable data, it faces inherent limitations, especially in urban environments characterized by complex landscapes with buildings, trees, and other structures. These obstacles create significant challenges in accurately estimating solar irradiance from satellite data alone, as the presence of urban features can obstruct the line of sight between the satellite and the ground, leading to errors in irradiance predictions. In urban areas, the interaction between solar irradiance and these structures results in a complex interplay of shadowing, reflection, and scattering, which satellite sensors may struggle to capture effectively¹⁹. This is particularly true for solar irradiance that reaches the ground in the form of direct sunlight, which can be blocked or reflected by tall buildings and other infrastructure. Furthermore, the diffuse component of solar radiation, which scatters through the atmosphere and reaches the ground from various angles, is often challenging to estimate accurately from satellite data due to the variability in cloud cover, atmospheric conditions, and local geography.

To address these limitations, the integration of sky camera data with satellite imagery has emerged as a powerful solution. Sky cameras provide real-time visual information of the sky, enabling the identification of cloud cover, obstructions, and other atmospheric conditions that can directly impact solar irradiance reaching the surface. Unlike geographic information systems (GIS) and other static spatial models to estimate solar irradiance²⁰, which rely on pre-existing maps and datasets, sky cameras offer continuous updates and can capture transient changes in the environment. This capability is particularly important because GIS-based models do not provide real-time information on newly constructed buildings²¹, temporary obstructions, or other structural changes that may alter solar exposure²². By incorporating sky camera data, solar irradiance estimation models can more accurately distinguish between direct and diffuse irradiance while also adapting to dynamic urban landscapes and evolving environmental conditions.

Recent research has investigated the integration of satellite remote sensing, ground-based sky imagery, and machine learning for solar irradiance estimation. For instance, Paletta et al.²³ combined satellite data and sky camera observations within a single machine learning framework to enhance intra-hour irradiance forecasting (up to 60 min ahead). Similarly, Rodríguez-Benítez et al.²⁴ integrated satellite and sky camera data to extend the temporal horizon of ASI-based nowcasting to 90 min. While these studies highlight the advantages of multi-source data fusion, they generally focus on fixed installations and do not address the specific challenges of estimating solar irradiance in dynamic urban environments. In contrast, our study focuses on irradiance estimation for moving vehicles, integrating satellite imagery, real-time sky camera data, and machine learning models.

In this study, we demonstrate the feasibility and accuracy of a novel model for estimating solar irradiance around moving vehicles. By integrating satellite remote sensing, sky camera observations, and machine learning

techniques, our proposed model significantly improves the reliability of solar irradiance estimation in complex urban environments. The contributions of this work can be summarized as follows:

1. We combine satellite imagery with ground-based sky camera observations to capture both direct and diffuse solar irradiance components with high temporal and spatial resolution.
2. Our model effectively fuses heterogeneous data sources, representing a significant improvement over conventional methods in accurately predicting solar irradiance.
3. We validate the methodology using real-world data collected from a vehicle equipped with solar irradiance sensors, temperature and humidity sensors, and GPS tracking, ensuring practical applicability.
4. The proposed methodology supports energy-aware climate control, enhances thermal comfort, and enables predictive modeling for both conventional and electric vehicles.

Data and instrument

Satellite data

This study primarily utilizes spectral satellite imagery from the GK2A satellite, which is equipped with an advanced meteorological imager (AMI)²⁵. The AMI features sixteen channels categorized into four groups: visible, near-infrared, mid-wave infrared, and long-wave infrared. These channels cover wavelengths ranging from 0.431 μm to 13.39 μm , with spatial resolutions varying between 0.5 km and 2.0 km. The GK2A/AMI observation data can be classified into three types based on spatial coverage: full-disk, extended local area, and local area. For this research, we utilized full-disk satellite data with a spatial resolution of 0.5 km and a temporal resolution of 10 min. Furthermore, the GK2A satellite level 2 dataset (https://www.data.go.kr/data/15139450/o_penapi.do?recommendDataYn=Y) offers 52 meteorological parameters, including temperature, humidity, and rainfall rate. We incorporate these data to model temperature and humidity under both stationary conditions and dynamic scenarios involving a moving EV.

Sensors on car

In this experiment, we utilized the IONIQ-5 electric vehicle, which was equipped with an array of sensors mounted on its roof, including pyranometer, relative humidity sensor, temperature sensor, GPS sensor, and sky camera, as illustrated in Fig. 1. These sensors were employed to capture a comprehensive range of environmental data, enabling a thorough understanding of the conditions surrounding the vehicle during the experiment. The sky camera used in this research was the USB3 Vision industrial camera U3-3280CP Rev.2.2, equipped with an FE185 series super wide-angle fish-eye lens. This fish eye lens provides a 185° field of view and is compatible with 5-megapixel high-resolution sensors, allowing the capture of wide-angle sky images with a resolution of 2272×1704 pixels, where each pixel value ranges from 0 to 255. Given that the vehicle's location varies with each experiment, precise positional data is essential for correlating with satellite information. Therefore, a GPS sensor was included to obtain accurate measurements of latitude, longitude, altitude, and the vehicle's direction. Further details regarding each sensor are provided in Table 1. Additionally, ground-based stationary data is incorporated for validation purposes. This dataset includes measurements of solar irradiance, ambient temperature, and humidity. The data is collected from a fixed monitoring station located at Kookmin University in Seoul, South Korea (Latitude: 37.611807, Longitude: 126.993986). To ensure temporal consistency between the mobile and satellite datasets, all sensor measurements from the vehicle were recorded at 10-minute intervals, matching the temporal resolution of the satellite data used in this study.

Study area

The experiment was conducted in Seoul, South Korea, a highly urbanized metropolis with a population of approximately 9.5 million and a total area of 605 square kilometers. As a city characterized by dense infrastructure and diverse environmental conditions, Seoul presents a challenging yet representative setting for solar irradiance prediction studies. Data collection was conducted near the Han River along a 7-kilometer route, as illustrated in Fig. 2. The route includes diverse urban environments such as high-rise structures, shaded zones, and areas with reflective surfaces. The experiment was conducted during winter and summer, a period of low and high solar

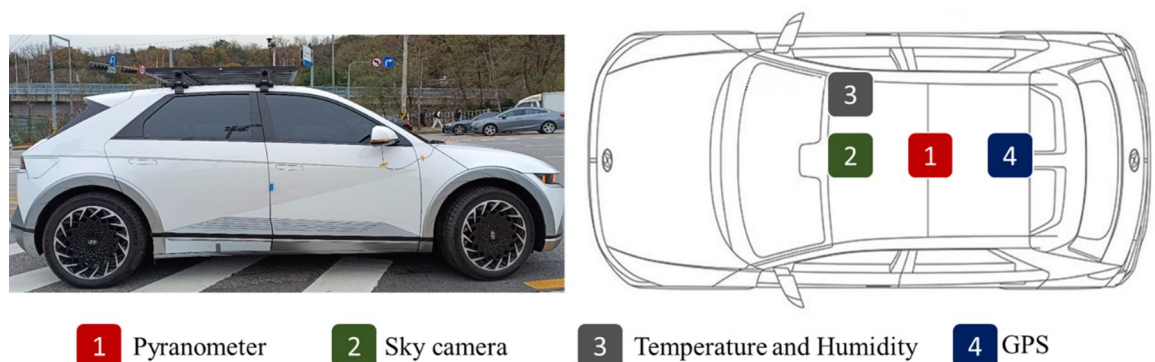


Fig. 1. Sensor locations and types installed on the IONIQ-5 EV.

Instrument type	Parameter	Specifications
EKO pyranometer ML-02	Solar irradiance	Measurement range: 0 to 2000 W/m ² Non-stability change/year: ± 2%
Sky camera	Sky image	Image resolution: 2448 × 2048 pixels
GPS16X-HVS	Location, speed, direction	Position accuracy: 95%
HygroVUE10	Temperature	Accuracy: ±0.2 °C Operating range: −40 to 70 °C
	Relative humidity	Accuracy: ±1.5% Operating range: 0 to 100%

Table 1. Parameters of sensors on EV.

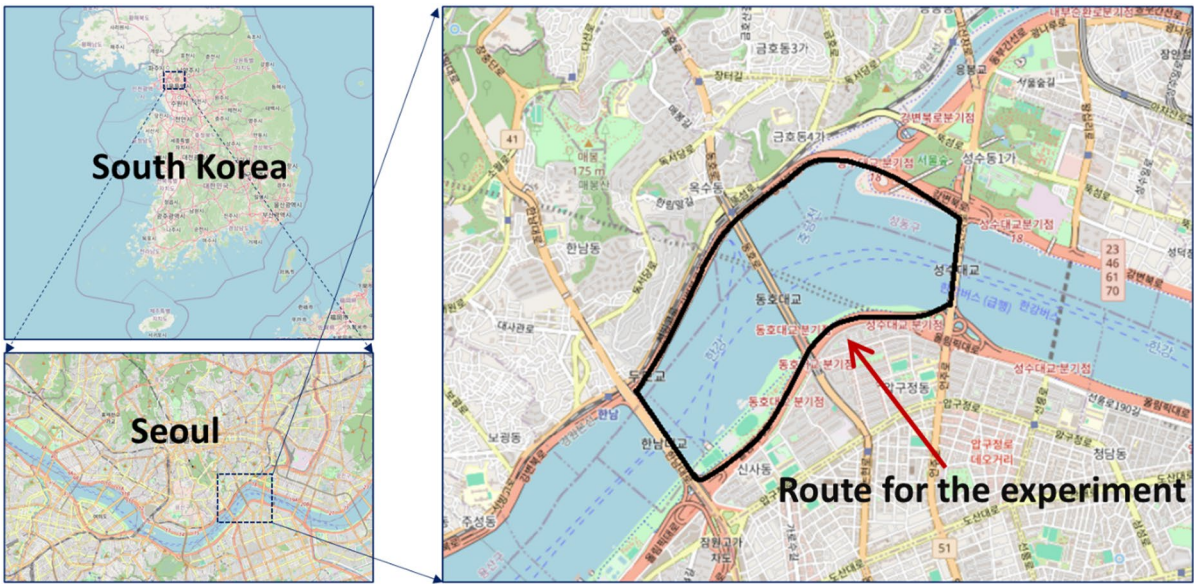


Fig. 2. Map of the experimental route in Seoul, South Korea. The map was created using QGIS²⁶, a free and open-source geographic information system. The black line represents the route followed during the field experiment for data collection.

intensity. This location was chosen to evaluate the model’s performance under realistic urban conditions where dynamic shading and weather variability significantly influence solar energy availability.

Methodology

In this section, the process of estimating solar irradiance around a moving EV is described, as illustrated in the flowchart in Fig. 3. The workflow begins with temperature and humidity estimation using GK2A Level-2 satellite data. It then incorporates multiple modeling approaches, including conventional methods such as semi-empirical and solar decomposition models, as well as machine learning techniques, to enable performance comparison and accuracy evaluation.

Data-driven model selection

The objective of this study is to estimate solar irradiance around a moving EV. To achieve this, three key models are required: (1) a satellite-based solar irradiance model to estimate global horizontal irradiance (GHI), (2) a solar irradiance decomposition model to derive diffuse horizontal irradiance (DHI) and direct normal irradiance (DNI), and (3) a sun-blocking calculation using sky image data. The satellite-based solar irradiance model, such as a semi-empirical approach^{27,28}, converts satellite image pixel values into GHI by separately accounting for atmospheric and cloud-related attenuation. These models are typically developed using data from the visible channels of geostationary satellites. The solar decomposition model, on the other hand, estimates DNI and DHI from GHI by utilizing the clearness index as the primary input and the diffuse fraction as the output. Among various solar decomposition models, the YANG5²⁹ model has demonstrated superior performance, outperforming its predecessors across 126 stations spanning diverse climate conditions. Further details on these methods are provided in the Appendix 1 and 2.

Although these models effectively estimate solar irradiance, they still have limitations. Since they rely on regression-based relationships derived from historical measurements, their accuracy can be affected by variations

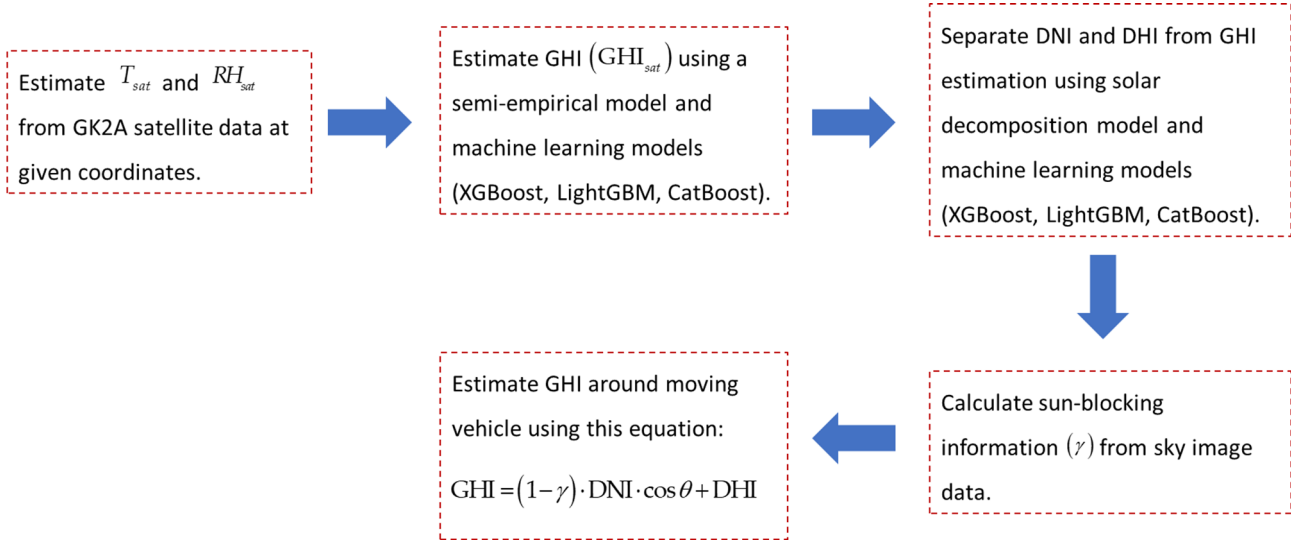


Fig. 3. Flow chart the process to estimate solar irradiance around moving vehicle in urban area.

Target	Input data
GHI	$P_{sat}, \theta, E_a, L_t, T_{sat}, RH_{sat}, AST, D_{es}$
DHI and DNI	$GHI_{sat}, \theta, E_a, L_t, T_{sat}, RH_{sat}, AST, D_{es}$

Table 2. The input data used for the machine learning model varies for each target variable.

in atmospheric conditions, cloud dynamics, and site-specific factors³⁰. Machine learning techniques offer a promising improvement by learning complex, nonlinear patterns from high-resolution satellite and ground-based data, potentially enhancing the robustness and adaptability of solar irradiance estimation, particularly for dynamic applications such as moving EVs. Therefore, we propose a data-driven machine learning approach to estimate solar irradiance components by leveraging the strengths of gradient-boosting frameworks, specifically tree-based machine learning models^{30,31}. In this study, we selected three models: extreme gradient boosting (XGBoost), light gradient boosting machine (LightGBM), and categorical boosting (CatBoost). Detailed explanations of these models are provided in Appendix 3. The input data for estimating GHI, DHI, and DNI are standardized based on the semi-empirical model and solar decomposition model. Additionally, temperature and humidity estimated from GK2A satellite data are also included, as presented in Table 2.

Unlike traditional linear regression models, which often struggle to accurately capture the complex relationships between input features and solar irradiance, tree-based model is well-suited for handling nonlinear dependencies and intricate interactions among variables. Tree-based model is an advanced ensemble learning algorithm based on decision trees, designed to optimize predictive performance while maintaining computational efficiency. Furthermore, its ability to handle missing data, prevent overfitting through regularization techniques, and efficiently process large datasets makes it particularly suitable for satellite-based solar irradiance estimation. Another key advantage of using tree-based-model is its compatibility with shapley additive explanations (SHAP)³², an interpretable machine learning technique that provides insights into feature importance. By applying SHAP analysis, we can quantify the contribution of each input variable to the model's predictions, helping to identify the most influential factors affecting solar irradiance. This enhances not only the transparency of the model but also its potential for further optimization by selecting the most relevant features.

Where P_{sat} represents the pixel value obtained from satellite imagery, θ is the solar zenith angle, E_a is the extraterrestrial irradiance, L_t corresponds to the turbidity-related parameter, T_{sat} is the air temperature estimated from satellite data, RH_{sat} is the satellite-derived relative humidity, AST denotes the apparent solar time, D_{es} is the distance between the sun and the Earth, and GHI_{sat} refers to the GHI estimated using satellite data.

Sky-image-based solar irradiance

Estimating solar irradiance in urban areas presents unique challenges, as it is affected not only by cloud cover but also by various physical obstructions, such as buildings, trees, and other structures, which reduce the solar irradiance reaching EV. Traditional methods, such as satellite data and decomposition models, often struggle to provide accurate estimations in these environments due to the complex interplay of shadows and sunlight. These methods are more suited for open areas, where obstacles are minimal. To improve accuracy in urban settings, advanced models capable of detecting shadows from obstructions are essential. Geographic information systems (GIS) have been proposed to identify and map these structures. However, GIS datasets are typically

static and frequently outdated, failing to capture real-time changes such as tree growth, new constructions, or temporary obstructions like parked vehicles or scaffolding. An alternative, more effective approach involves the use of sky cameras for solar irradiance estimation. Sky cameras offer continuous, real-time monitoring of the environment, capturing dynamic changes and detecting whether the sun is partially or fully obscured by obstacles. This approach is highly beneficial in complex urban landscapes, where shadows shift rapidly due to the changing position of the sun and the movement of objects. Real-time image data provides a much more accurate assessment of solar irradiance, particularly in areas where traditional methods falter.

The key concept is to determine whether the sun is obstructed and to calculate the proportion of the sun being blocked using image processing techniques, as illustrated in Fig. 4. This process begins by converting sky images into single-channel images, which enhances contrast and reduces noise, using the red-blue ratio (RBR) method. This step makes it easier to identify the sun, sky, clouds, and any obstructions. As outlined in Eq. 14, the proportion of the sun that is blocked (γ) is computed through this image processing. In clear sky conditions, when the sun is completely obstructed by objects, only the DHI is considered for calculating the GHI.

$$\text{GHI} = (1 - \gamma) \cdot \text{DNI} \cdot \cos \theta + \text{DHI} \quad (1)$$

Evaluation metrics

The accuracy of the estimation models is assessed through multiple statistical metrics, including root-mean-square error (RMSE) and mean bias error (MBE). To enable meaningful comparisons across diverse sky conditions, normalized metrics such as normalized RMSE (nRMSE) and normalized MBE (nMBE) are also employed. These normalized values are computed by dividing each metric by the mean of the actual measured values.

$$\text{MBE} = \frac{1}{n} \sum_{i=1}^n (y_e - y_m) \quad (2)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_e - y_m)^2} \quad (3)$$

where y_m , y_e , $\overline{y_m}$, n represent the actual data, estimated data, mean value of actual data, and the number of samples, respectively.

Results and discussion

Results validation using stationary data

The estimation model is first validated under stationary conditions before being applied to moving conditions. This validation consists of two parts: environmental conditions, including temperature and relative humidity, estimated using level 2 GK2A satellite data, and solar irradiance components estimated using machine learning models. Figure 5 presents scatterplots comparing the estimated temperature and relative humidity from GK2A level 2 satellite data with ground-based measurements. The validation results indicate nRMSE of 23% for temperature, with a MBE of -1.24°C , suggesting a slight underestimation. The nRMSE and nMBE for relative humidity are 27.17% and 15.73%, respectively, highlight the challenges in accurately capturing humidity variations, likely due to atmospheric complexities affecting satellite retrieval process.

Table 3 provides a comprehensive summary of the solar irradiance estimation results obtained using various tree-based machine learning models. In this analysis, two years of historical data were employed for training the models, followed by one year of data for testing to evaluate their performance. The findings clearly indicate that the machine learning models significantly surpass the accuracy of both the semi-empirical model and the solar decomposition model in predicting solar irradiance. Among the machine learning models evaluated, the

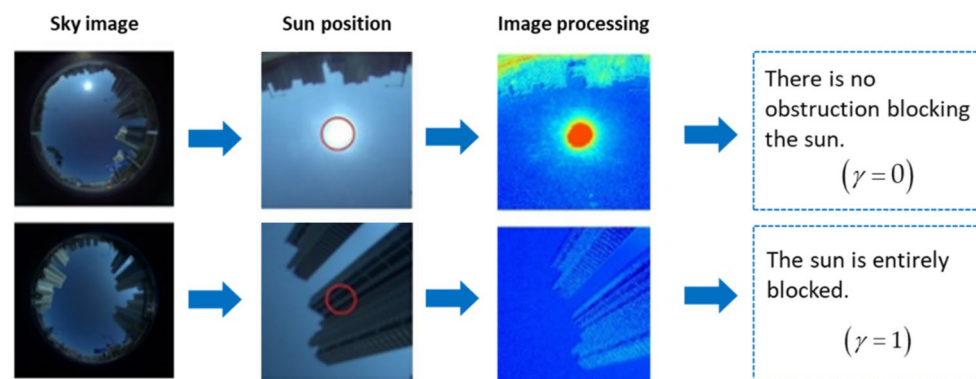


Fig. 4. Sky image analysis to detect shadows caused by obstructions such as buildings, street light poles, trees, and other structures, with a focus on calculating the proportion of the sun's light blocked by these obstructions.

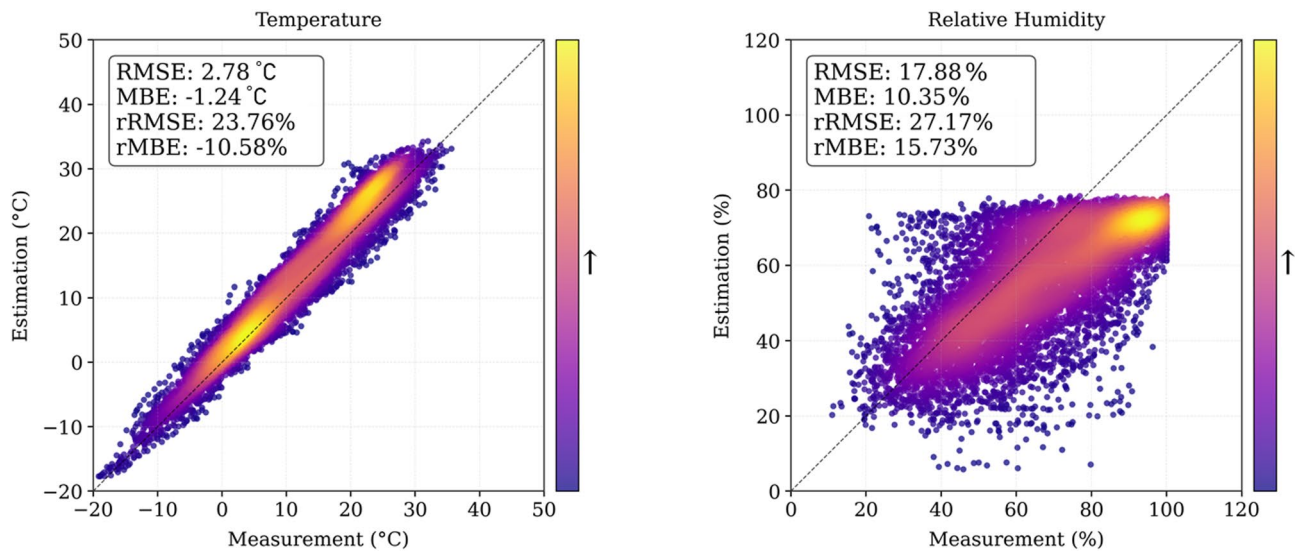


Fig. 5. Scatter plots of temperature and relative humidity estimation using satellite GK2A.

	GHI		DNI		DHI	
	nRMSE (%)	nMBE (%)	nRMSE (%)	nMBE (%)	nRMSE (%)	nMBE (%)
XGBoost	18.11	−3.97	30.21	−2.67	30.95	4.15
LightGBM	16.23	−3.55	28.54	−1.55	29.52	2.44
CatBoost	14.55	−0.21	27.99	−0.27	27.61	1.28
Semi-empirical	23.54	−8.92	-	-	-	-
Solar decomposition	-	-	33.43	−11.4	32.57	8.77

Table 3. Results of estimated solar irradiance using all machine learning models.

CatBoost model stands out for its exceptional performance. It achieves the highest accuracy in estimating all key components of solar irradiance, including GHI, DNI, and DHI. The superior accuracy of the CatBoost model suggests that its ability to handle complex, non-linear relationships in the data is particularly effective for this application. The consistent and reliable performance of the CatBoost model, led us to adopt it as our preferred model for estimating solar irradiance around moving EV.

Figure 6 shows the results of estimated solar irradiance for each component with nRMSE of 14.55%, 27.99%, and 29.78% for GHI, DNI, and DHI, respectively using CatBoost model. The scatter plots demonstrate the relationship between measured and predicted values across all three irradiance components during the stationary validation phase. For GHI estimation, the CatBoost model achieves exceptional accuracy (nRMSE: 14.55%, RMSE: 58.06 W/m², nMBE: −2.1%) by leveraging geostationary satellite (GK2A) pixel data as primary input features. The DNI component shows moderately higher scatter (nRMSE: 27.99%, RMSE: 75.41 W/m²) but maintains excellent overall balance with minimal systematic bias (nMBE: −0.2%). Similarly, the DHI component demonstrates slightly higher relative error (nRMSE: 29.78%, RMSE: 63.93 W/m²) with a small positive bias (nMBE: 4.65%). The increased scatter in both components reflects the inherent challenges in modeling directional and diffuse radiation under variable atmospheric conditions. These comprehensive validation results confirm the model’s reliability in accurately estimating all three irradiance components under stationary conditions before application to the more complex dynamic environment of moving EVs.

Figure 7 presents SHAP summary plots for three types of solar irradiance: GHI, DNI, and DHI using CatBoost model. Each subplot visualizes the impact of different input features on the model predictions, with features listed on the y-axis and SHAP values on the x-axis. The SHAP values indicate the contribution of each feature to the model’s prediction, with positive values increasing the predicted irradiance and negative values decreasing it. The color gradient represents the magnitude of each feature value, transitioning from purple (low values) to yellow (high values). The vertical spread of points along the y-axis for each feature reflects the distribution of that feature across all samples. A wider spread indicates greater variability in the feature within the dataset, showing that different values of the same feature contribute differently to the prediction. Conversely, a narrow spread reflects lower variability in the feature values across samples.

From the plots, certain features have a stronger influence on the predictions, as indicated by the larger spread of SHAP values. For instance, in all three plots, solar zenith angle and extraterrestrial irradiance appear to be among the most influential features, as they show a broad range of SHAP values. Other meteorological variables like temperature and relative humidity also contribute but with lower SHAP magnitudes. Additionally, the

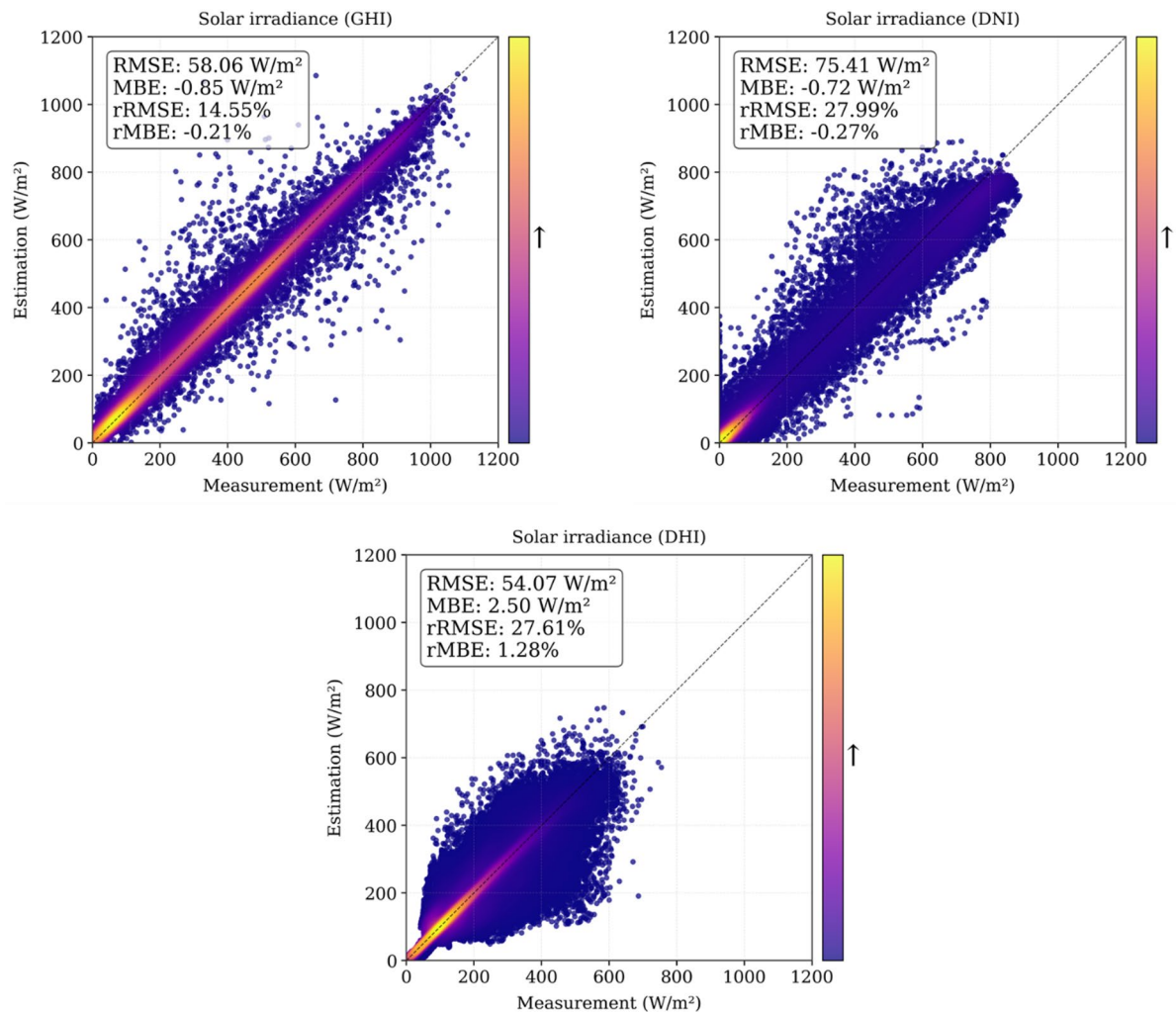


Fig. 6. Solar irradiance estimation using CatBoost model: GHI (left), DNI (middle), and DHI (right). The color gradient indicates the density of data points, with yellow representing the highest concentration in the scatter plot.

ordering of feature importance varies slightly between GHI, DNI, and DHI, indicating that each component of solar irradiance is influenced differently by these factors. The black arrows on each plot suggest an increasing trend in the feature's influence, highlighting how certain variables have a more significant impact as their values increase.

This analysis is particularly important for the context of solar irradiance estimation in a moving vehicle, where environmental and positional factors change dynamically. To the best of our knowledge, previous studies have not systematically explored the relative importance of different input parameters for estimating GHI, DNI, and DHI in such moving vehicles. Therefore, the SHAP analysis is not only useful for interpreting the machine learning model but also provides novel insights into which input variables most significantly affect each type of irradiance under real-world driving conditions in urban environments.

Solar irradiance on urban areas

In open spaces, the GHI derived from satellite data can be directly assumed to represent the GHI incident on an EV. However, in urban environments, this assumption is no longer valid due to various obstructions, such as buildings, which can significantly reduce the solar irradiance reaching the EV. These obstructions primarily block direct irradiance, making it crucial to determine whether the EV is shaded or exposed. The most accurate and up-to-date approach for obtaining this information is through real-time imaging. In this context, a sky camera serves as an optimal tool, as it can detect surrounding structures while simultaneously capturing the sky conditions, including the sun's position.

Figure 8 presents an example of sun obstruction detection, illustrating three scenarios: fully blocked, partially blocked, and unblocked sunlight. In these cases, the sun is identified as the brightest region in the image, with pixel values at the maximum intensity near its expected position. The sky camera captures images in three color channels (red, green, and blue), each with pixel values ranging from 0 to 255. To enhance contrast and minimize noise, the images are converted to grayscale using the red-blue ratio (RBR) method. This transformation

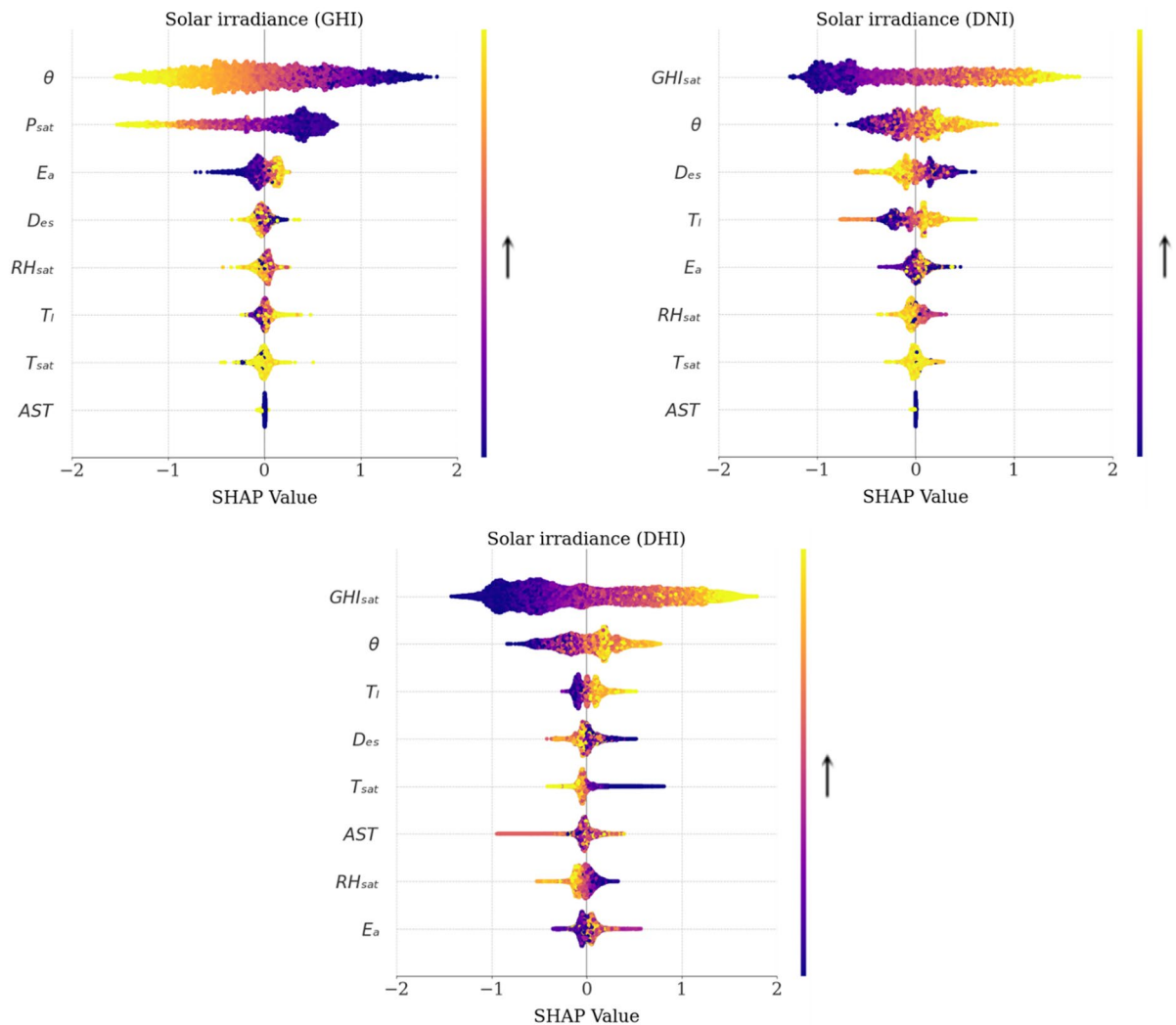


Fig. 7. SHAP values displaying the effect of different features on the solar irradiance estimation, including GHI, DNI, and DHI. The y-axis shows the individual input features ranked by their impact, while the x-axis represents the magnitude of the SHAP values. The horizontal bars indicate each feature contribution to the model prediction.

facilitates the accurate classification of sun, cloud, and sky pixels. The sun's radius is predefined as 50 pixels, and the sun is identified as the region containing the highest pixel intensity (255). The proportion of the sun's obstruction is then computed based on brightness distribution. This information is essential for quantifying the direct irradiance reaching the EV, which can subsequently be used to refine solar irradiance calculations using Eq. 1. It should be noted that, in the absence of direct measurement data for sun obstruction, validation is initially performed through human observation. Once the relevant data are collected, they can be converted into estimated solar irradiance and subsequently compared with the corresponding measured values for validation.

Solar irradiance Estimation around the moving EV

Figures 9 and 10 present the estimated and measured solar irradiance values for a moving EV in an urban environment during winter and summer, respectively. The plots include GHI, DNI, and DHI, with an additional parameter, γ , representing sun blockage due to buildings or other structures. A high γ value indicates complete obstruction of sunlight, while lower values signify partial or no blockage. Seasonal differences in solar irradiance characteristics are evident, with winter showing lower irradiance levels due to reduced solar altitude and shorter daylight hours, whereas summer exhibits higher irradiance, more significant fluctuations, and stronger DNI contributions.

In winter, GHI remains consistently low and stable, while DNI experiences occasional peaks due to temporary clear-sky conditions. The higher DHI values indicate frequent cloud cover and urban obstructions diffusing solar radiation. Periods with γ close to 1 align with significant declines in DNI, suggesting considerable solar blockage from buildings. The estimation model faces challenges under these conditions, leading to discrepancies between estimated and measured GHI. The winter results show an nRMSE ranging from 17% to 23%, with a negative

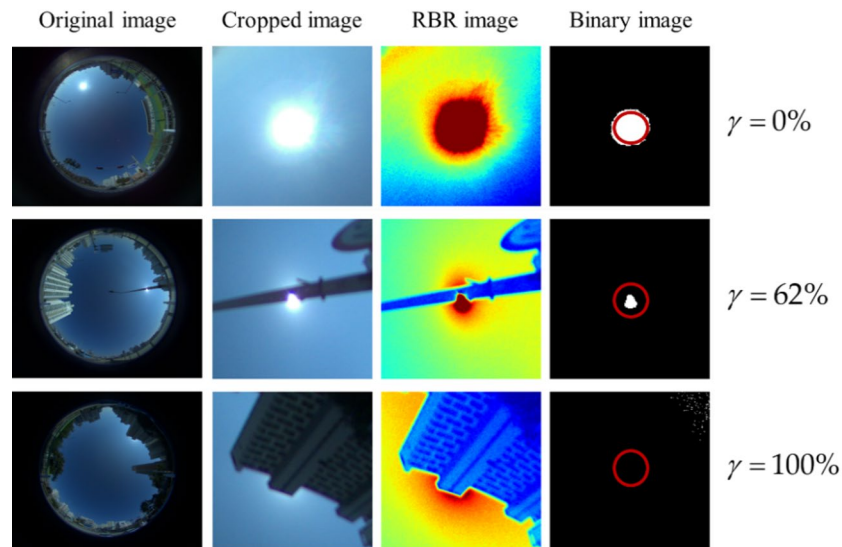


Fig. 8. The process of determining the percentage of sun obstruction by structures involves analyzing sky images in three scenarios: fully blocked, partially blocked, and completely unobstructed sun.

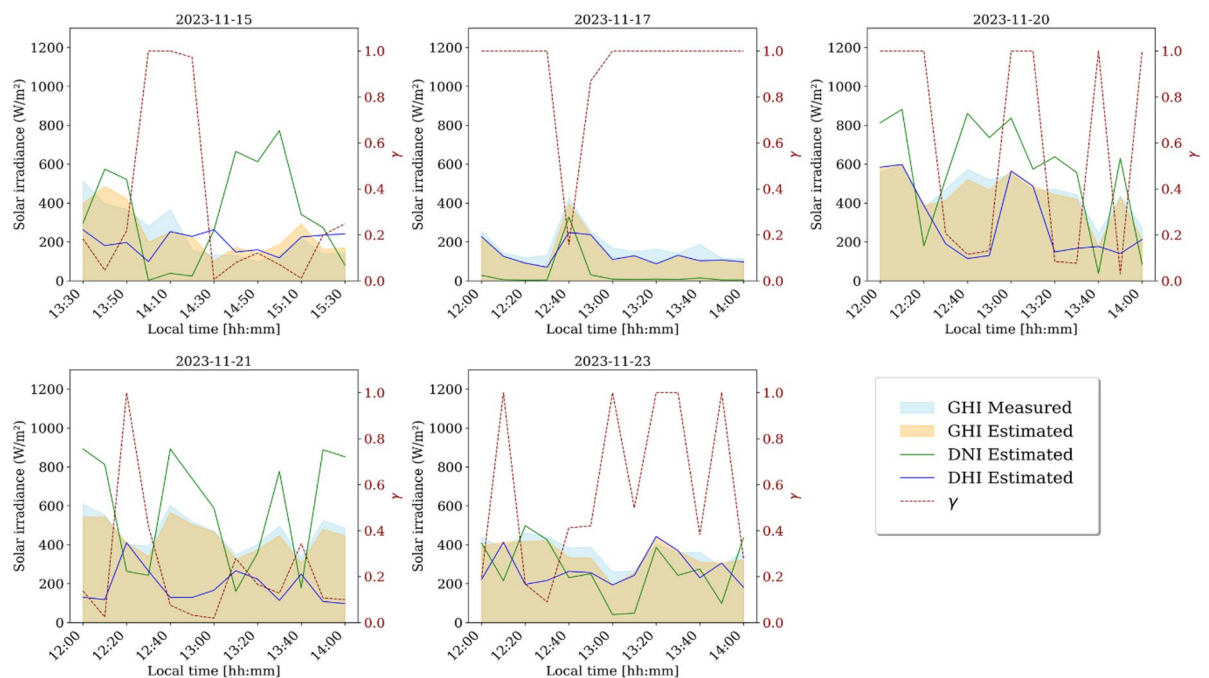


Fig. 9. Solar irradiance estimations around a moving EV during a 5-day winter experiment.

MBE as shown in Table 4, which indicates that the model tends to underestimate solar irradiance, particularly during periods of high solar blockage, causing a consistent bias in the estimations.

In summer (Fig. 9), GHI values are generally higher, with DNI showing more frequent peaks, which correspond to clearer sky conditions and less cloud cover. However, fluctuations in γ reveal intermittent solar blockage caused by buildings and structures, which introduces variability in both DNI and DHI measurements. These fluctuations contribute to the complexity of accurately estimating solar irradiance in an urban environment. Despite these challenges, the estimation model performs more effectively during the summer months, as the presence of higher irradiance. The best results for this season achieve an nRMSE of 14.61%. However, similar to the winter season, the MBE remains negative as shown in Table 4, indicating a slight underestimation of solar irradiance.

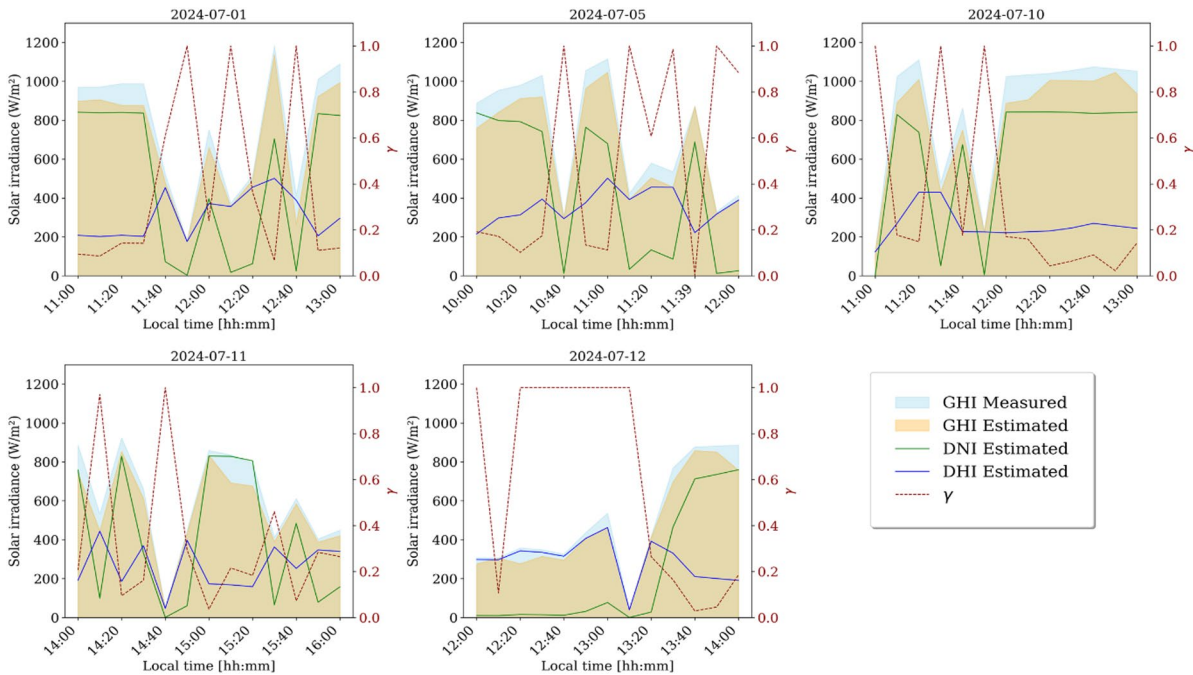


Fig. 10. Solar irradiance estimations around a moving EV during a 5-day summer experiment.

Winter					
	2023-11-15	2023-11-17	2023-11-20	2023-11-21	2023-11-23
RMSE (W/m ²)	65.42	42.91	84.89	83.50	78.58
MBE (W/m ²)	−21.31	−34.94	−20.66	−30.10	−23.06
nRMSE (%)	17.10	23.52	18.45	17.67	21.23
nMBE (%)	−5.57	−19.15	−4.49	−6.37	−6.23
Summer					
	2024-07-01	2024-07-05	2024-07-10	2024-07-11	2024-07-12
RMSE (W/m ²)	111.87	118.67	131.41	104.93	83.68
MBE (W/m ²)	−70.14	−62.51	−73.96	−57.03	−41.64
nRMSE (%)	14.61	16.27	15.28	17.35	16.76
nMBE (%)	−9.16	−8.57	−8.60	−9.43	−8.34

Table 4. Results of the solar irradiance estimation on moving EV during winter and summer season.

Conclusion

This study presents a comprehensive methodology for real-time solar irradiance estimation around a moving vehicle by integrating geostationary satellite imagery (GK2A), ground-based sky images, and machine learning algorithms. Our data-driven approach was validated through a two-stage process that confirms both its scientific validity and practical utility. The stationary validation phase demonstrated exceptional accuracy using the CatBoost model, with nRMSE values of 14.55% for GHI, 27.99% for DNI, and 27.61% for DHI, while maintaining low bias with nMBE of −0.21%, −0.27%, and 1.28%, respectively. When applied to a moving vehicle in urban environments, our approach maintained robust performance with nRMSE values below 24% in winter and below 17% in summer.

A key innovation is the use of sky image analysis to detect real-time obstructions, including urban infrastructure, foliage, transportation elements, and transient objects that block direct solar radiation. Unlike static GIS-based methods that rely on outdated structural information and cannot account for dynamic environmental factors, our system continuously adapts to the actual surroundings of the vehicle. This real-time obstruction detection enables smart HVAC control strategies that can accurately respond to changing solar conditions, significantly reducing unnecessary energy consumption while maintaining optimal passenger comfort.

Although thermal management is a 24-hour concern, our method specifically addresses the significant and highly variable impact of solar irradiance during daytime driving, which can increase cooling loads and reduce energy efficiency if not properly managed. Unlike static measures such as improved insulation or high-performance glazing, the proposed dynamic, location-aware system adjusts HVAC operation in real time according to changing environmental conditions, enhancing responsiveness and efficiency during daylight

hours. Nevertheless, the model has some limitations: it slightly underestimates temperatures in winter due to low solar angles and increased atmospheric attenuation, and real-time implementation requires sufficient onboard computational resources, while accessing satellite data for real-time operation requires a stable internet connection. Geographic scalability also depends on satellite coverage; while GK2A provides comprehensive coverage across East Asia, extending the system to other regions would require integration with alternative geostationary satellites.

Future work will explore predictive capabilities to forecast solar exposure along vehicle routes, enabling proactive HVAC control. Integration with on-board predictive control systems and vehicle digital twins could allow coordinated optimization of thermal management alongside other energy-consuming systems in smart mobility frameworks. Combining this approach with complementary strategies, such as optimized insulation and nighttime thermal regulation, can further enhance energy efficiency throughout the diurnal cycle, supporting sustainable and comfortable mobility solutions.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Appendix

Appendix-1

Semi-empirical solar irradiance model

A semi-empirical model is used to transform pixel values from satellite images into GHI, considering the impact of atmospheric and cloud-related attenuation separately. Semi-empirical models are typically developed to exploit data recorded by the visible channel of a geostationary satellite.

In the initial stage of this process, the pixel value is adjusted by normalizing it, considering factors such as air mass, the distance between the Sun and Earth, and the solar elevation.

$$\rho = \frac{P_{sat} \cdot A_m \cdot D_{es}}{(2.238h^{-0.26} \cdot \exp(-0.004h))} \quad (4)$$

where P_{sat} is pixel value from satellite image, A_m is air mass, D_{es} is the Sun-Earth distance (AU), and h is solar elevation. Then, the cloud index (CI) value is determined by:

$$CI = \frac{\rho - \rho_{low}}{\rho_{high} - \rho_{low}} \quad (5)$$

where ρ_{low} and ρ_{high} are the lowest and highest of normalized pixels.

The next step is to convert CI to clear sky index, k_{cs} , and then to GHI, using the Hammer model²⁷ in Equation 6. Finally, the GHI is computed by employing the Ineichen-Perez³³ clear sky, GHI_{cs} , model with turbidity estimates derived from local meteorological measurements.

$$k_{cs_Hammer} = \begin{cases} 1.2; & CI \leq -0.2; \\ 1.0 - CI; & -0.2 < CI \leq -0.8; \\ 2.0667 - 3.6667 \cdot CI + 1.6667 \cdot CI^2; & 0.8 < CI \leq 1.1; \\ 0.05; & 1.1 < CI; \end{cases} \quad (6)$$

$$GHI_{Hammer} = GHI_{cs} \cdot k_{cs_Hammer} \quad (7)$$

The Hammer model was selected due to its superior performance compared to other models³⁴.

Appendix-2

Solar decomposition model

Direct and diffuse irradiance are separated from GHI using the solar decomposition model. The solar decomposition model is a nonlinear regression model that utilizes the clearness index as its primary input and the diffuse fraction as its output. The YANG5 model is the most recent of the numerous solar decomposition models that have been developed²⁹. It has been tested at 126 stations across a broad range of climate types and has demonstrated superior accuracy.

$$k_d^{YANG5} = C + \frac{1 - C}{1 + e^{\beta_0 + \beta_1 k_t + \beta_2 AST + \beta_3 \theta + \beta_4 \Delta k_{tc} + \beta_5 k_{d, \text{hourly}}^{ENERGER2} + \beta_6 k_{de}}} + \beta_5 k_{de} \quad (8)$$

$$k_d = \frac{DHI}{GHI} \quad (9)$$

$$k_t = \frac{GHI}{E_a \cos \theta} \quad (10)$$

$$k_{de} = \max \left(0, 1 - \frac{\text{GHI}_{cs}}{\text{GHI}} \right) \quad (11)$$

$$\Delta k_{tc} = \frac{\text{GHI}_{cs}}{E_a} - k_t \quad (12)$$

where θ is solar zenith angle, GHI_{cs} is clear-sky irradiance, E_a is extraterrestrial irradiance, AST is apparent solar time, k_d is diffuse fraction, k_t is clearness index, $k_{d,\text{hourly}}^{\text{ENERGER2}}$ is hourly diffuse fraction using ENGERER2 model¹⁴, and $C, \beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6$ are coefficients based on the regime²⁹.

Appendix-3

Tree-based machine learning models

Extreme gradient boosting (XGBoost)

XGBoost is an optimized implementation of the Gradient Boosting framework, designed to be highly efficient, flexible, and portable³⁵. It builds decision trees sequentially, where each new tree attempts to correct the errors of the previous ones. The core idea is to minimize a custom regularized objective function, which balances model complexity and predictive power, making it less prone to overfitting. XGBoost introduces techniques like column subsampling, tree pruning, and parallel processing to enhance performance. It also supports handling missing values internally, making it a popular choice for structured/tabular data tasks like solar irradiance estimation. One of XGBoost's key features is its use of second-order Taylor expansion to optimize the loss function, which enables precise updates and efficient learning. It also includes regularization parameters that help control overfitting, a critical aspect when dealing with complex datasets. The use of histogram-based optimization further speeds up tree construction by grouping continuous data into discrete bins. However, due to its complexity and memory usage, XGBoost can sometimes be slower for very large datasets compared to its competitors.

Light gradient boosting machine (LightGBM)

LightGBM is a gradient boosting framework designed to handle large datasets efficiently³⁶. Unlike XGBoost, which grows trees level-wise (depth-first), LightGBM uses a leaf-wise growth strategy, where it expands the leaf with the maximum loss reduction. This approach leads to deeper trees with potentially higher accuracy. LightGBM also employs a histogram-based decision tree algorithm, which significantly reduces memory usage and speeds up computation, making it well-suited for large-scale solar irradiance data. A notable feature of LightGBM is its use of gradient-based one-side sampling (GOSS) and exclusive feature bundling (EFB). GOSS selectively uses data instances with larger gradients to train trees, improving efficiency without sacrificing accuracy. EFB reduces dimensionality by bundling mutually exclusive features, allowing LightGBM to handle high-dimensional data more effectively than XGBoost. However, the aggressive leaf-wise approach can sometimes lead to overfitting if not properly regularized.

Gradient boosting with categorical features support (CatBoost)

CatBoost is a gradient boosting algorithm designed to handle categorical features directly without the need for one-hot encoding, which is a significant advantage for datasets with many categorical variables³⁷. It employs an ordered boosting technique that reduces prediction bias by using permutations of the data during training. This approach helps prevent overfitting, making CatBoost a strong candidate for solar irradiance estimation tasks that might involve categorical data like cloud types or sensor locations. Another key feature of CatBoost is its ability to combat the gradient bias caused by target leakage using a novel method called ordered boosting. Unlike XGBoost and LightGBM, which can suffer from overfitting with categorical data, CatBoost's native handling of these features ensures better generalization. While CatBoost is generally slower to train than LightGBM, it often achieves superior accuracy due to its advanced handling of categorical features and mitigation of overfitting risks.

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Author contributions

RAR: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. HSL: Conceptualization, Funding acquisition. HJL: Conceptualization, Writing – review & editing, Supervision, Formal analysis, Funding acquisition.

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Declarations

Competing interests

The authors declare no competing interests.

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